

General criteria to consider before buying an MCU board for prototyping

Serviceability. It is important for the boards to be repairable when something goes wrong. Packaging of the MCU needs to be considered too. It is important to examine if the MCU is a through-hole part-mounted in a socket or a surface-mounted one. If it is surface-mounted, it is important to know if it can be replaced when the need arises.

Pedagogy. The educational value and cognitive loading effects of the board must be taken into account. Big and fancy development boards may offer tremendous number of features and look really cool, but if these features impact a user's understanding of the core technology, then these may be detrimental to the cause.

Reliability. One of the main goals of a development board is to ensure an operational development board for each and every experiment. Any time spent attempting to fix the development board can take away a lot of critical and important time.

Programming support/options. Newer MCUs allow in-circuit programming while these are located in the system. Options may include an external programmer or through a self-program mechanism. Short-term and long-term consideration should be taken in account. If the development board is utilised for other activities, these should be considered when deciding how the board will be programmed.

Processor support (single or multiple). If one is working on multiple controller families or different part numbers, this factor becomes critical, as having a separate development board for each controller may not be feasible all the time. A proper decision can save money in the short term as well as help in efficiently developing products or projects in the long run.

Operating voltage. Many controllers these days support multi-voltage operation and circuits could have multiple sections operating at different voltages such as 1.8V, 3.3V, 5V, 12V or so on. Therefore development tools must support multi-voltages along with level converters, wherever needed.

Cost. It is an important factor. Irrespective of the value an MCU board provides to the user, the cost should be at par.

Design control. There is nothing more frustrating than selecting an appropriate development board only to have the manufacturer make a change that ultimately forces a new development board to be selected. It is therefore necessary to have some type of design control over the development board that is ultimately chosen.

Power supply. Users would not want to buy another power supply just to suit a certain type of development board. There

are a multitude of options for supplying power to a development board including benchtop supplies, wall warts, batteries or via USB connections. As long as the development board provides some kind of support on power supply management (whether the board has a rectifier, switched mode power supply (SMPS) or whole power supply on the board), its value is retained.

Debugging capabilities. Depending upon the kind of work to be done on a development board, Joint Test Action Group (JTAG), single-wire debug or any other functionality is necessary so as to enable users do easy debugging and faster development.

System clock options. This gives the board multiple clock options such as an internal resistor-capacitor (RC) oscillator and/or an external crystal to be installed, which can make or break some of the developments. This is not very critical when compared to all other factors, though.

Serial port/USB connectivity. For many years, RS232 serial port connections between development boards and computers have been a norm. However, laptops with RS232 ports are almost impossible to find nowadays. Recent trends have led to USB serial ports as the primary connection between boards. These may be with native USB support or in the form of serial-to-USB converters. Either way, this option is critical to development.

Additionally, one may want more than one communication port (COM). One of the ports can be used in project development, while the other can be used for debugging and other purposes. Therefore dual- or multiple-connectivity options are ideal to have.

Form factor. While development boards are ultimately used for development purposes, their form factor is critical. A board, which is much bigger in size and difficult to move around may not be preferred, while a small-sized handy board, which can be taken anywhere in a pocket or a small box could help engineers develop things any time.

Readiness for final integration. Buyers also need to look for embeddable boards, such that if they finalise a solution and time-to-market is short, development boards could be directly integrated into the solution. In many cases, development boards become a perfect fit for the final product. In such cases, designing another board may not be warranted (except when parameters like outline of board, form factor, etc are different). This becomes a very important factor since it has high potential to save not only time but also redesign and manufacturing costs.

Courtesy: T. Anand, managing director, Knewron

tal displays (LCDs), thin-film transistor (TFT) displays and communication ports with onboard transceivers are popular among users. "Many users also look for onboard sensors, which may be as simple as the ones for sensing the X-Y-Z coordinate and as complicated as the ones that perform medical applications like pulse oximetry," notes Nishant Bhaskar, embedded processing application engineer, Texas Instruments India.

He adds, "In case of power electronics based systems, customers not only look for the extent to which the power stage has been designed but

also the configurability on offer, if users want to tweak it."

Another very important feature in present-day development boards from companies like STMicroelectronics and TI is onboard debugging, informs Virupaksha. He says, "Apart from beginners, most users require this feature. In fact, it is a boon for developers, as they need not invest in a debugger, which costs approximately ₹ 8000."

Even with all these new features, Thushar feels, basic I/O pins, universal synchronous asynchronous receiver transmitters (USARTs), serial

peripheral interface (SPI), I2C, two-wire serial interface (TWI), timers, compare/capture/pulse width modulation (CCP), analogue-to-digital converters (ADCs) and digital-to-analogue converters (DACs) constitute basic features of an MCU, which should always be accounted for as far as a development board is concerned.

With increased focus on smart-phone based connected applications, consumers also prefer to work with wireless MCU kits [specifically in the Wi-Fi/Bluetooth Low Energy (BLE)/Bluetooth domain], which have some kind of Android or iOS app to interact